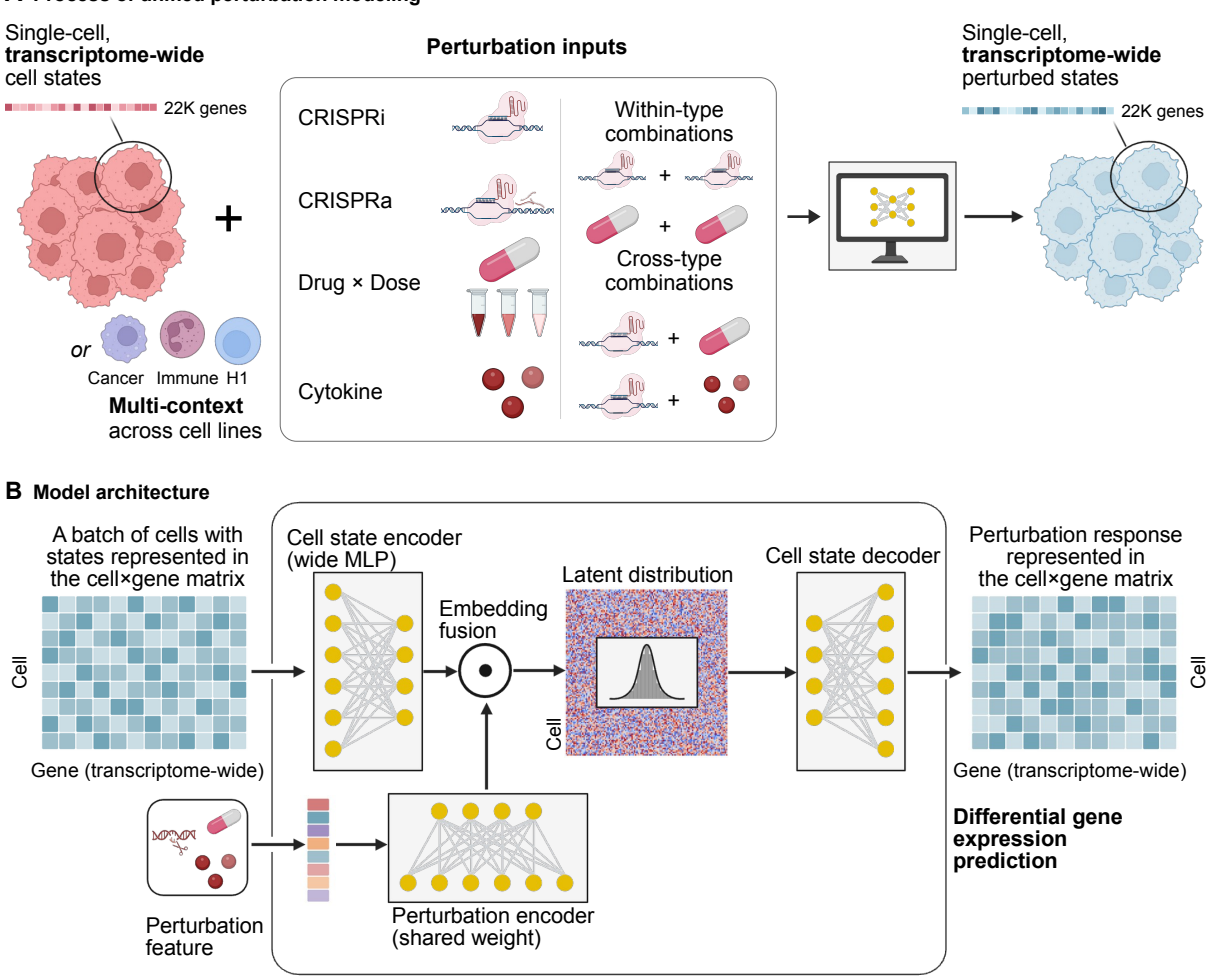

A Process of unified perturbation modeling
Single-cell, transcriptome-wide cell states
22K genes
or
Cancer
Immune
H1
Multi-context across cell lines
Perturbation inputs
CRISPRi
CRISPRa
Drug × Dose
Cytokine
Within-type combinations
Cross-type combinations
Single-cell, transcriptome-wide perturbed states
22K genes
B Model architecture
A batch of cells with states represented in the cell×gene matrix
Cell
Gene (transcriptome-wide)
Cell state encoder (wide MLP)
Embedding fusion
Latent distribution
Cell
Cell state decoder
Perturbation response represented in the cell×gene matrix
Cell
Gene (transcriptome-wide)
Differential gene expression prediction
Perturbation feature
Perturbation encoder (shared weight)